WORD EMBEDDING BASED SEARCH ENGINE

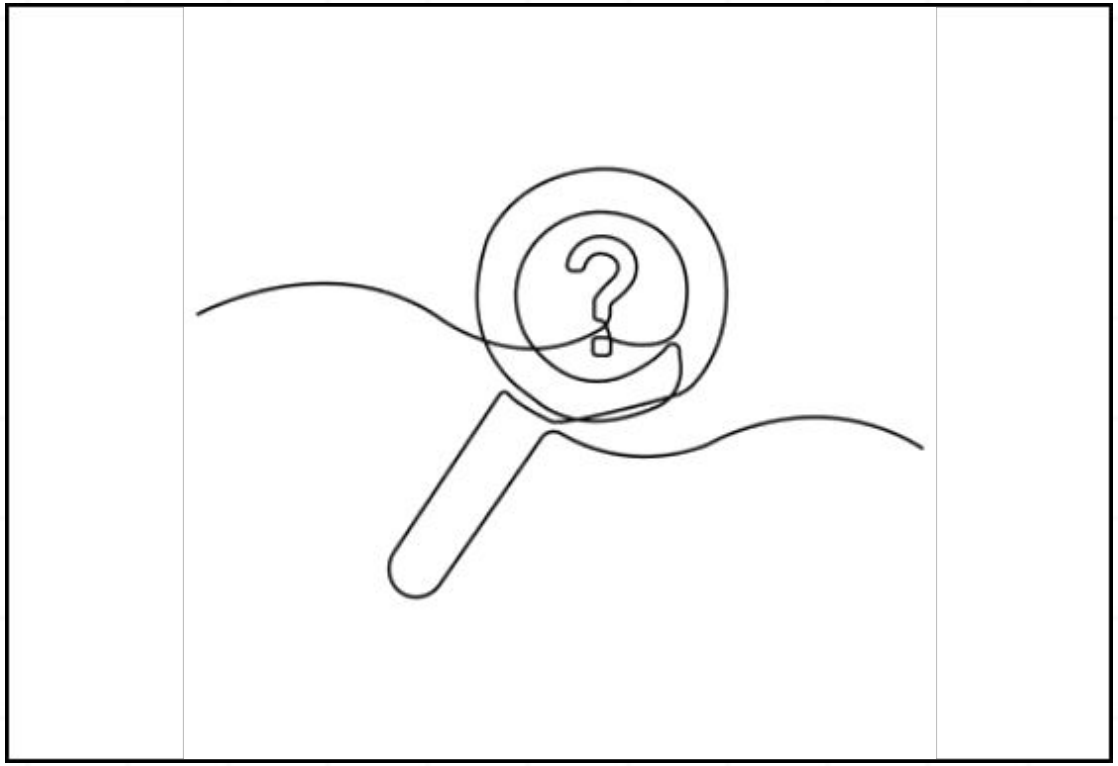
E-Commerce Data Science Project

OBJECTIVE & PROBLEM STATEMENT

Transitioning from rigid, keyword-based mapping to fluid, context-aware semantic retrieval.

INTRODUCTION TO SEMANTIC SEARCH

- **Keyword Limitations:** Traditional search fails to handle natural language variations, resulting in incomplete and irrelevant matches.
- **Contextual Meaning:** Semantic search retrieves products based on intended meaning rather than strict character sequences.
- **Vector Representation:** By utilizing advanced transformer models (Sentence-BERT), text is plotted into a continuous mathematical space.
- **Cosine Proximity:** Relevance is determined by calculating the angular distance between a query vector and product vectors.



DATASET DESCRIPTION

SOURCE MATERIAL

The system utilizes the **Flipkart Products Dataset**, containing over 20,000 e-commerce product listings across multiple departments.

Raw fields include product_category_tree, description, product_specifications, and retail_price.

UNIFIED SCHEMA

Data is parsed into a strict 18-column schema to normalize inputs for the models. Key parsed fields include:

- Department, Category, Subcategory
- Features, Color, Material, Size Range
- Price, Rating, and Review Counts

https://www.kaggle.com/datasets/atharvjairath/flipkart-ecommerce-dataset

DATASET HANDLING & PREPROCESSING

- **Cleaning Text Fields:** The preprocessor module strips all HTML tags, removes stray URLs, and normalizes unicode characters.
- **Boilerplate Removal:** Frequently occurring e-commerce phrases (e.g., “30 day replacement”, “cash on delivery”) are eliminated using regex to prevent bias in embeddings.
- **Hinglish Translation:** Common colloquial terms are mapped to English equivalents (e.g., "sasta" → "cheap", "kapde" → "clothes") to improve semantic understanding.
- **Deduplication:** Exact duplicates (based on product ID) and near-duplicates (same name and category) are removed to maintain dataset integrity.
- **Normalization:** Numerical attributes such as price and rating are standardized, with outliers handled via percentile capping and value clipping. Ratings are strictly clipped between 0.0 and 5.0.

DATA ENRICHMENT PIPELINE



TEMPLATE ENGINE

enricher.py dynamically generates 300+ word descriptions based on department-specific formatting templates.



METADATA BINDING

Scattered JSON fields of product attributes like color, material, and category are combined into meaningful, easy-to-read descriptions.



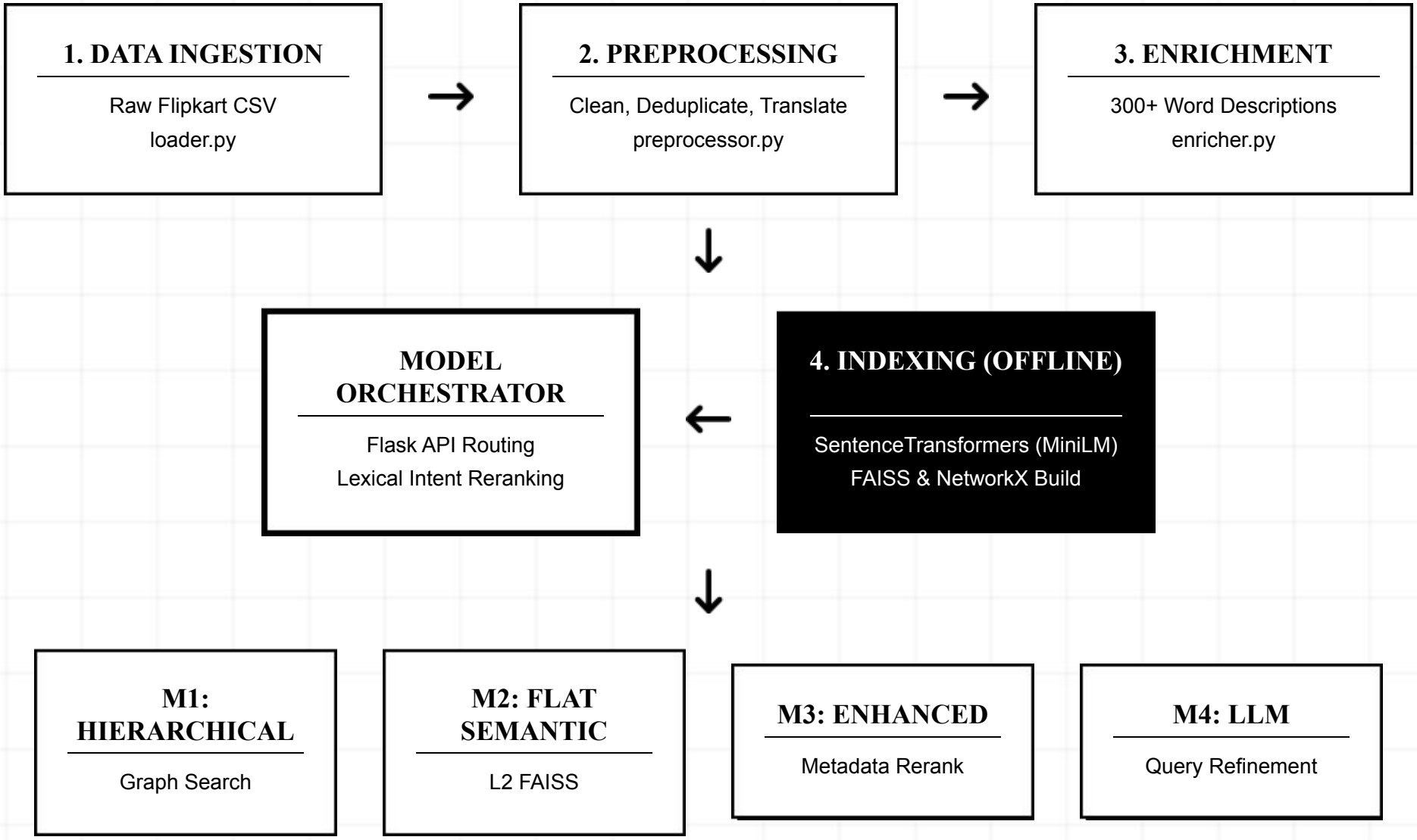
SEMANTIC DEPTH

Adding more contextual detail helps the model better understand products and improves the quality of embeddings.

SYSTEM ARCHITECTURE

A robust multi-model orchestrator powered by Flask, FAISS, and NLP

SYSTEM ARCHITECTURE FLOW



EMBEDDING STRATEGY & FAISS

384

Vector Dimensions

ALL-MINILM-L12-V2

The SentenceTransformer model maps product descriptions into a dense 384-dimensional space. The framework is optimized for CPU efficiency with offline batch processing capability.

FAISS INDEXING RULES

Utilizes FlatL2 (exact nearest neighbor) for datasets under 50,000 products ensuring maximum recall. It calculates angular distance to convert proximity into a normalized 0-1 confidence score.

MODEL 1: HIERARCHICAL GRAPH SEARCH

ARCHITECTURE LOGIC

Operates on a 5-level taxonomy logic implemented via a directed acyclic graph (DAG) in **NetworkX**:

- ROOT
- Department
- Category
- Subcategory
- Product

SCORE PROPAGATION

Embeddings are generated for taxonomy names at each level. Queries are compared at every node.

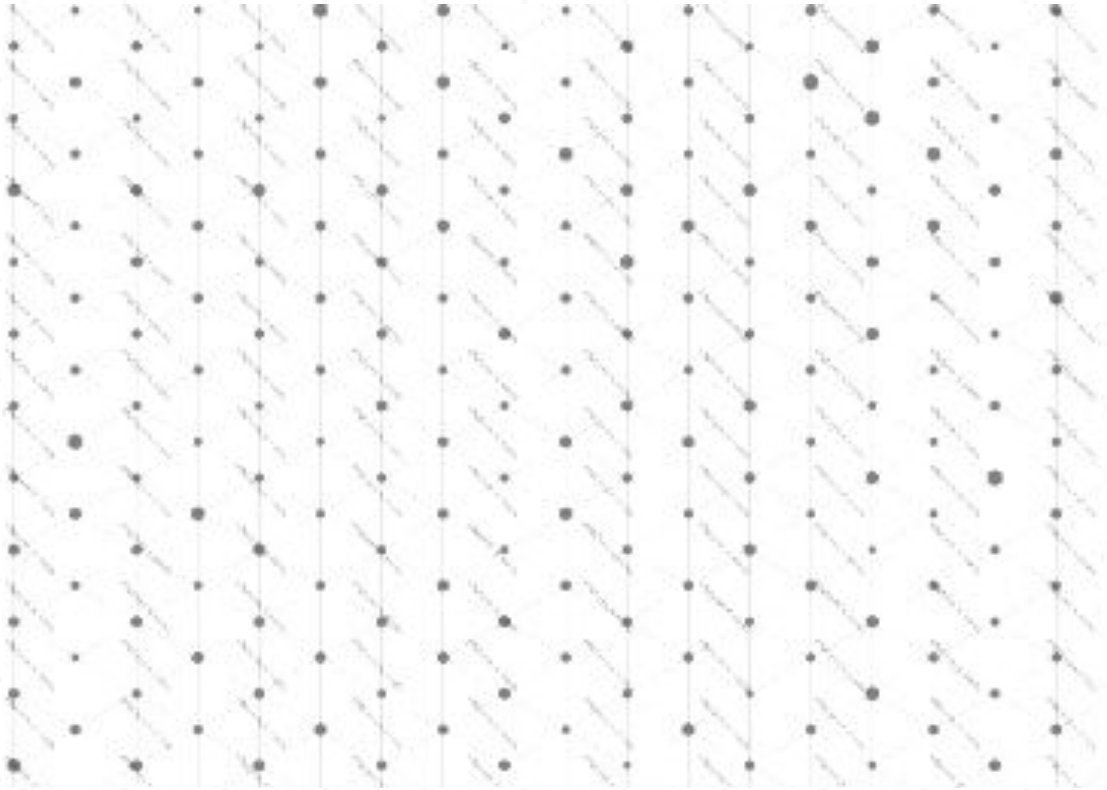
A strict decay factor dictates final relevance, preventing low-level node matches from overriding poor high-level category matches.

$$S = \frac{S_1\gamma^4 + S_2\gamma^3 + \ \ldots\ + S_5}{5}$$

MODEL 2: FLAT SEMANTIC SEARCH

Treats the entire product catalog uniformly, ignoring strict taxonomic boundaries for broader cross-category discovery.

- **Feature Extraction:** Combines bare-minimum cleaned text fields (Name, Category, Basic Features) into a single string.
- **Vectorization:** Uses Sentence-BERT to generate a single representation vector per product.
- **FAISS Retrieval:** Searches the entire flat vector space using exact L2 distance mapping.
- **Confidence Mapping:** L2 distance is inversely scaled into a 0.0 to 1.0 confidence score probability.



MODEL 3: ENHANCED SEMANTIC SEARCH

ENRICHED EMBEDDINGS

Extends Model 2 by embedding the 300+ word contextual outputs generated by enricher.py, rather than basic concatenated strings. Over-fetches 5x candidates via FAISS to allow for metadata re-ranking.

METADATA RE-RANKING WEIGHTS

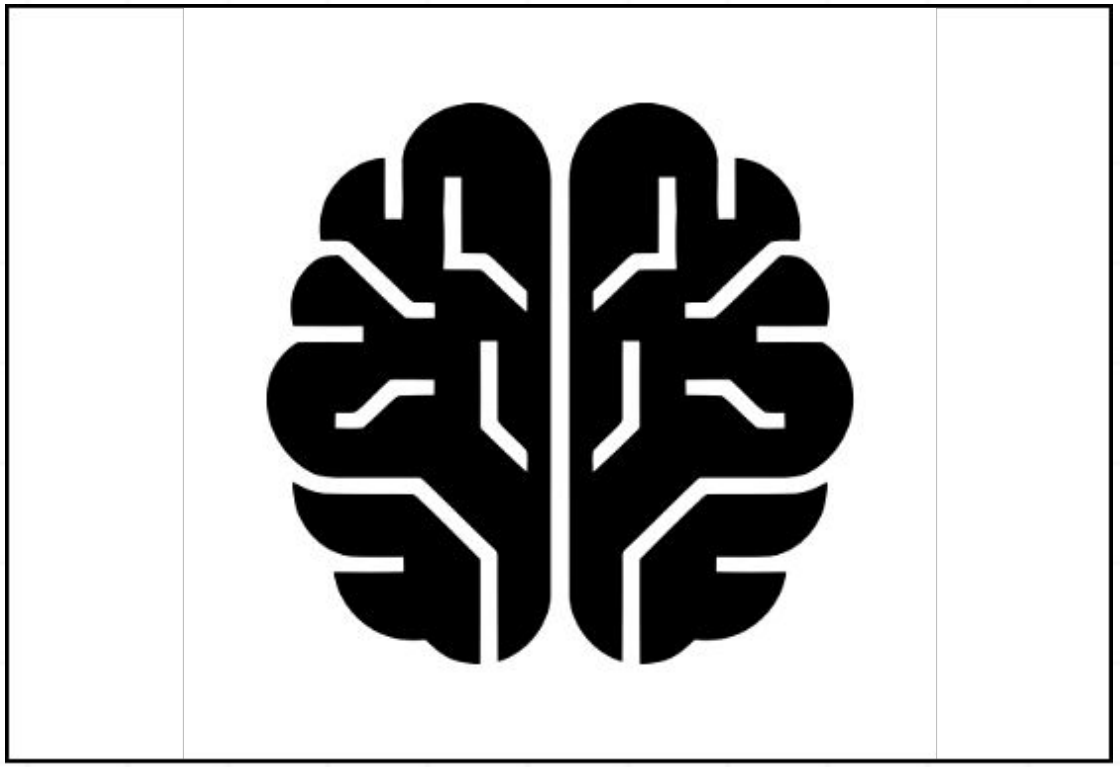
Applies a deterministic formula incorporating product quality signals:

- **Semantic Similarity:** 85%
- **User Rating:** 5% (Normalized 0-1)
- **Review Count:** 2% (Log scale capped)
- **Price Relevance:** 3% (Validity signal)
- **Stock Status Bonus:** 5%

MODEL 4: LLM-ENHANCED SEARCH

Solves the "conversational query" problem where users ask for intent rather than nouns.

- **OpenAI Interception:** Vague queries are passed to gpt-3.5-turbo with a strict system prompt.
- **Vocabulary Expansion:** The LLM translates intent (e.g., "keep money") to precise nouns (e.g., "wallet purse cardholder").
- **Model 3 Pass-Through:** The expanded query is then executed against the Model 3 FAISS index.
- **Deterministic Fallback:** If the API hits rate limits or fails, a regex-based intent dictionary expands the query safely.



SEARCH ORCHESTRATOR & ENSEMBLE



LEXICAL FILTERS

The Orchestrator extracts strict intent constraints via regex (e.g., detecting "mens", "kids") before vector search.



SCORE ADJUSTMENT

Products violating extracted lexical constraints receive severe penalty multipliers (0.35x), forcing alignment with explicit intent.



WEIGHTED VOTING

The Ensemble executes all 4 models and merges results based on weight: Model 3 (35%), Model 4 (25%), Model 1 & 2 (20% each).

DEMO & EVALUATION

Live Query Outputs and Metric Benchmarks

E-Commerce Semantic Search

Find products using natural language — not just keywords

Try: 'comfortable shoes for office work' or 'something to read books digitally'

Search

Model: Auto (Model 3 — Best Balance)

Results: Top 5

Try these queries:

comfortable office shoes

athletic footwear for running

device for reading books

something to watch movies

kitchen tools for cooking

workout clothes

LIVE QUERY EXECUTION

INSPECTING SEARCH RESULTS FOR A QUERY

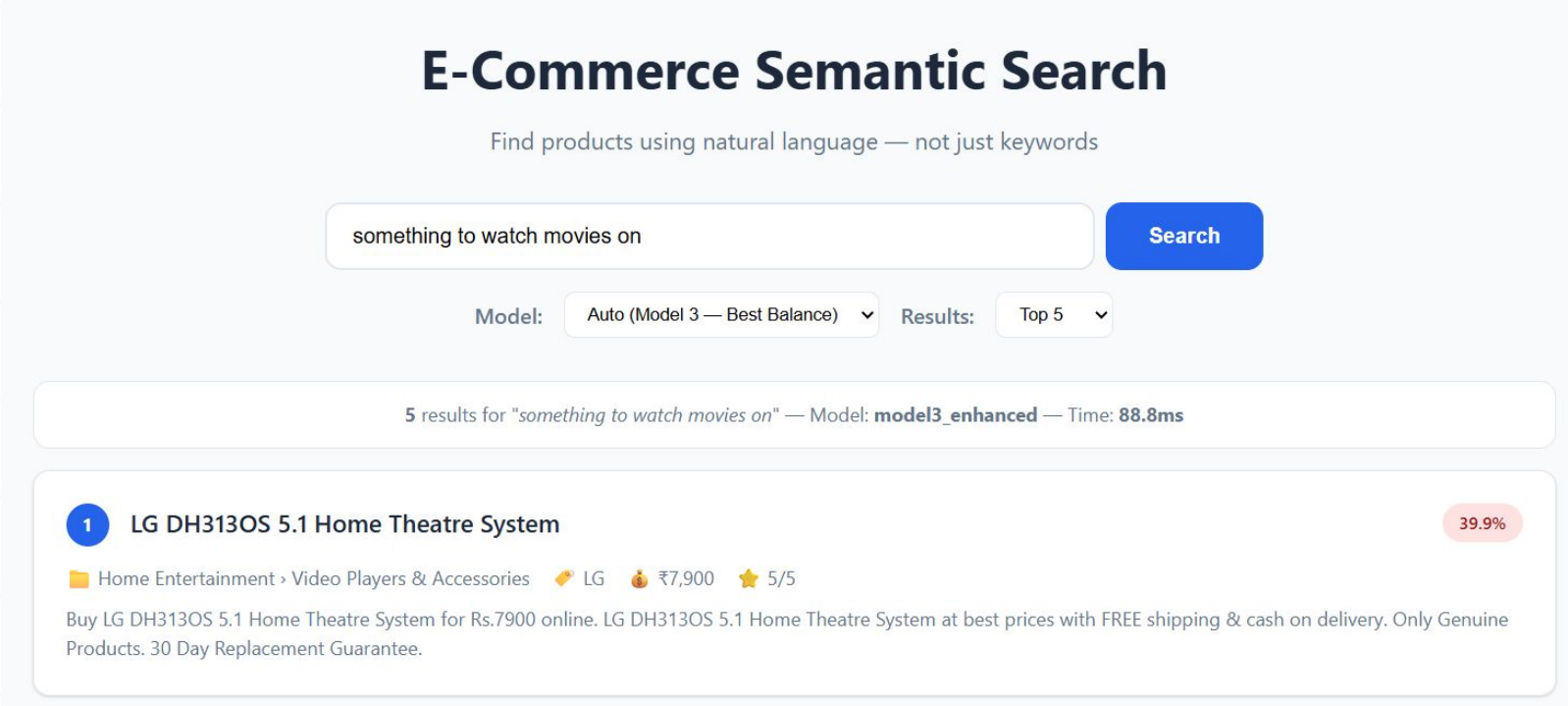
Input Query:

"something to watch movies on"

Intent Category: Vague / Conversational

Ground Truth Term Matches:

- Television / TV
- Projector
- Home Theatre / Monitor



QUERY RESULTS MODEL 1

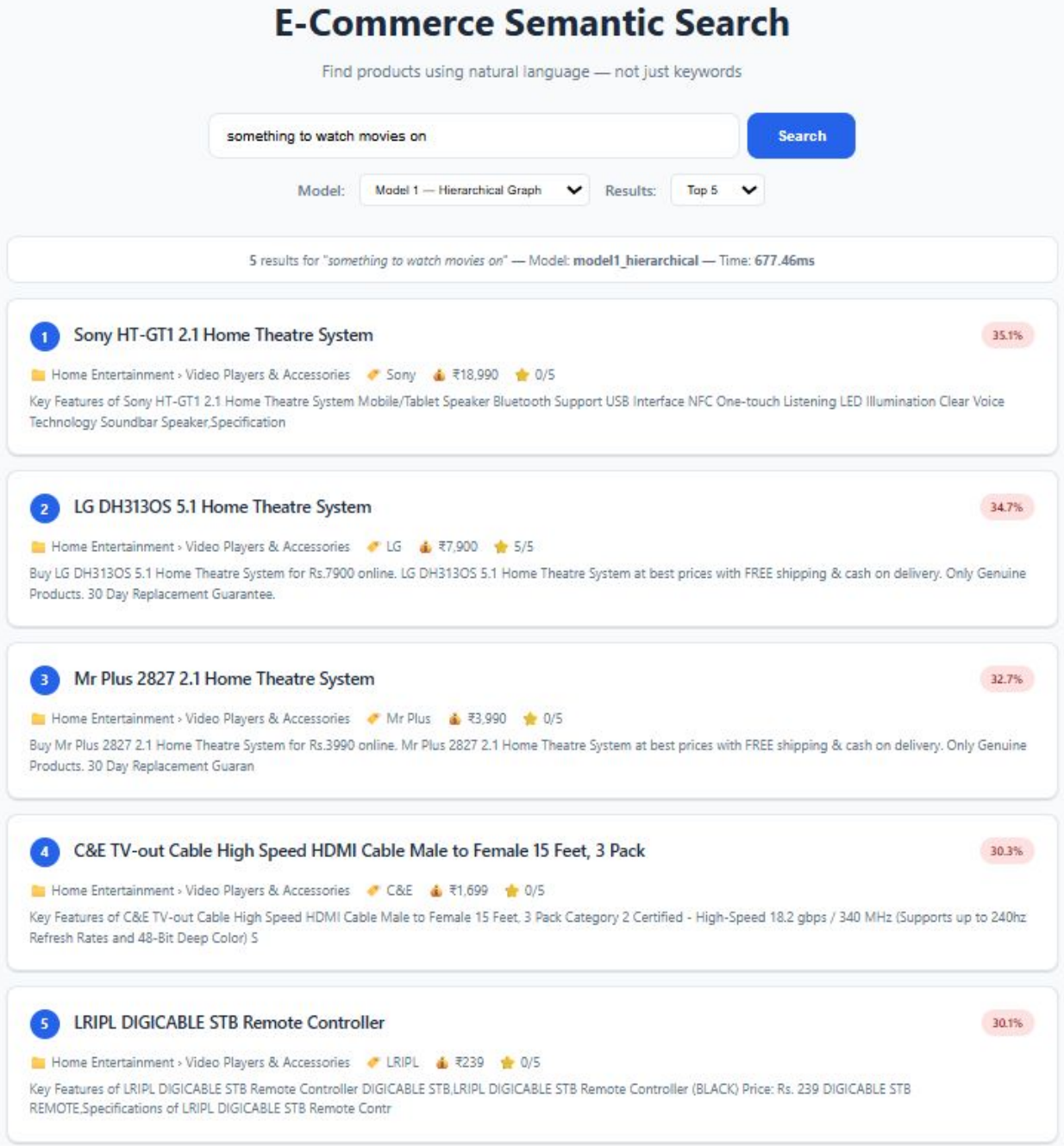
Query: "something to watch movies on"

- Maps the query to the **Home Entertainment category**, so all results are structurally consistent but limited to that branch.

- Retrieves **home theatre systems (matches ground truth)** but also irrelevant items like HDMI cables and remotes.

- Fails to return key ground truth items like **TV, projector, and monitor**, showing weak understanding of user intent.

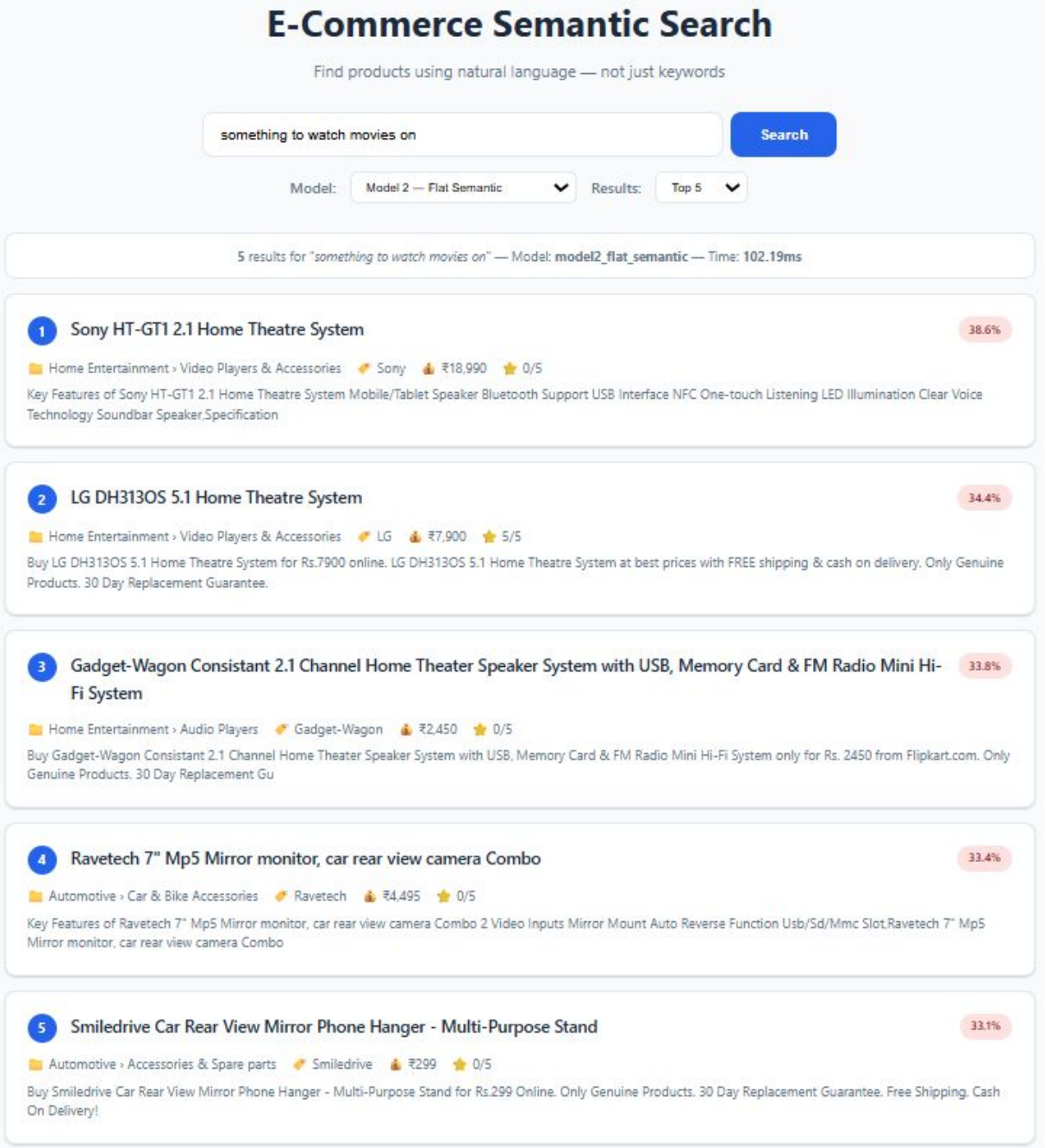
- Low similarity scores (~30–35%) due to **taxonomy-based matching with score decay**, where vague queries lead to generic category-level results.



QUERY RESULTS MODEL 2

Query: "something to watch movies on"

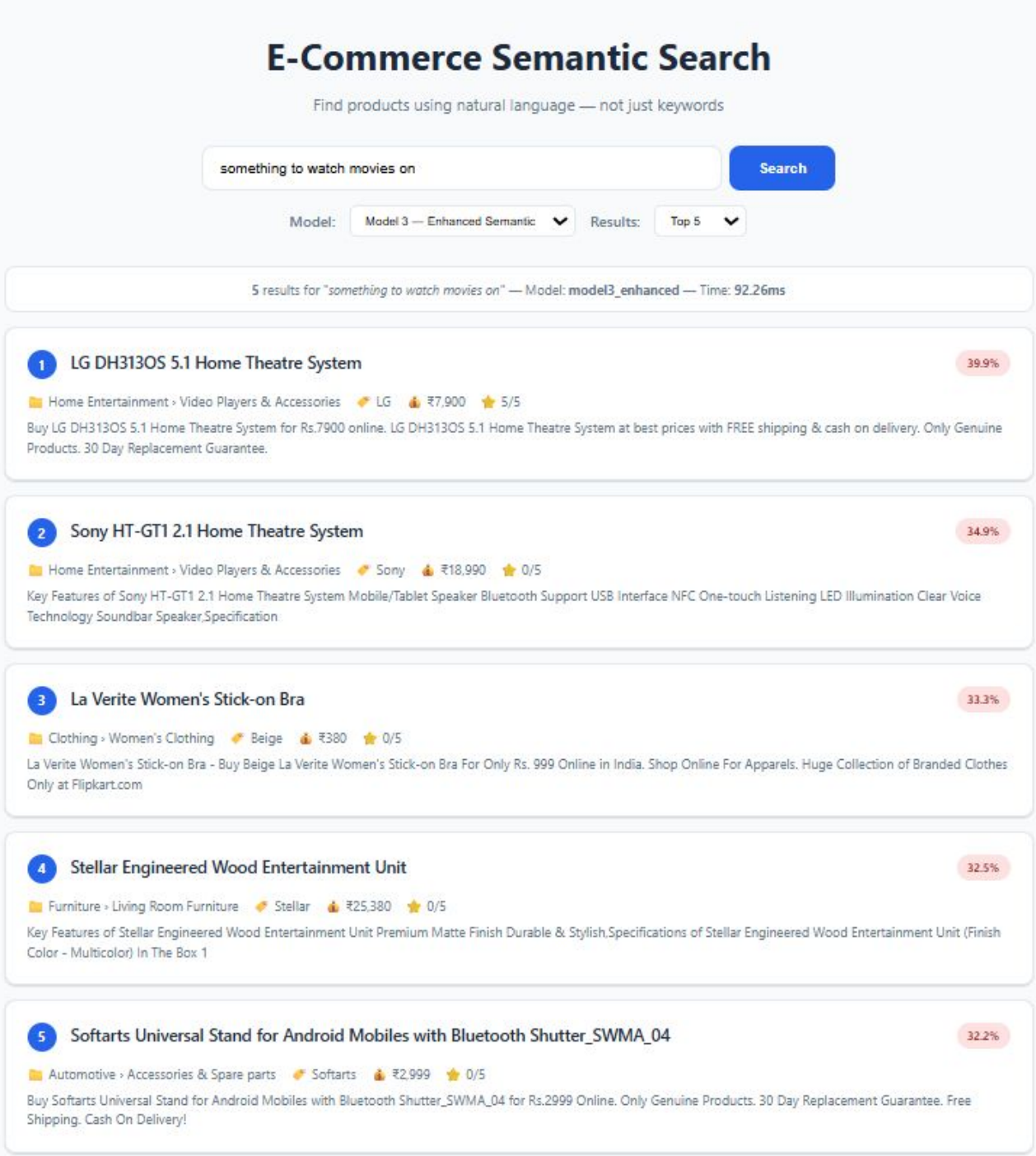
- Results are still dominated by home theatre systems, which match the intent (movie watching) but also miss key ground truth items like TV or projector.
- Unlike Model 1, it also retrieves a monitor and unrelated automotive items, showing that removing hierarchy allows broader but noisier results.
- The model captures semantic meaning (“watch movies”) better than strict hierarchy, but lacks filtering, so irrelevant items (car accessories) appear.
- Faster response and slightly higher top score, but trade-off is reduced precision due to no category constraints.



QUERY RESULTS MODEL 3

Query: "something to watch movies on"

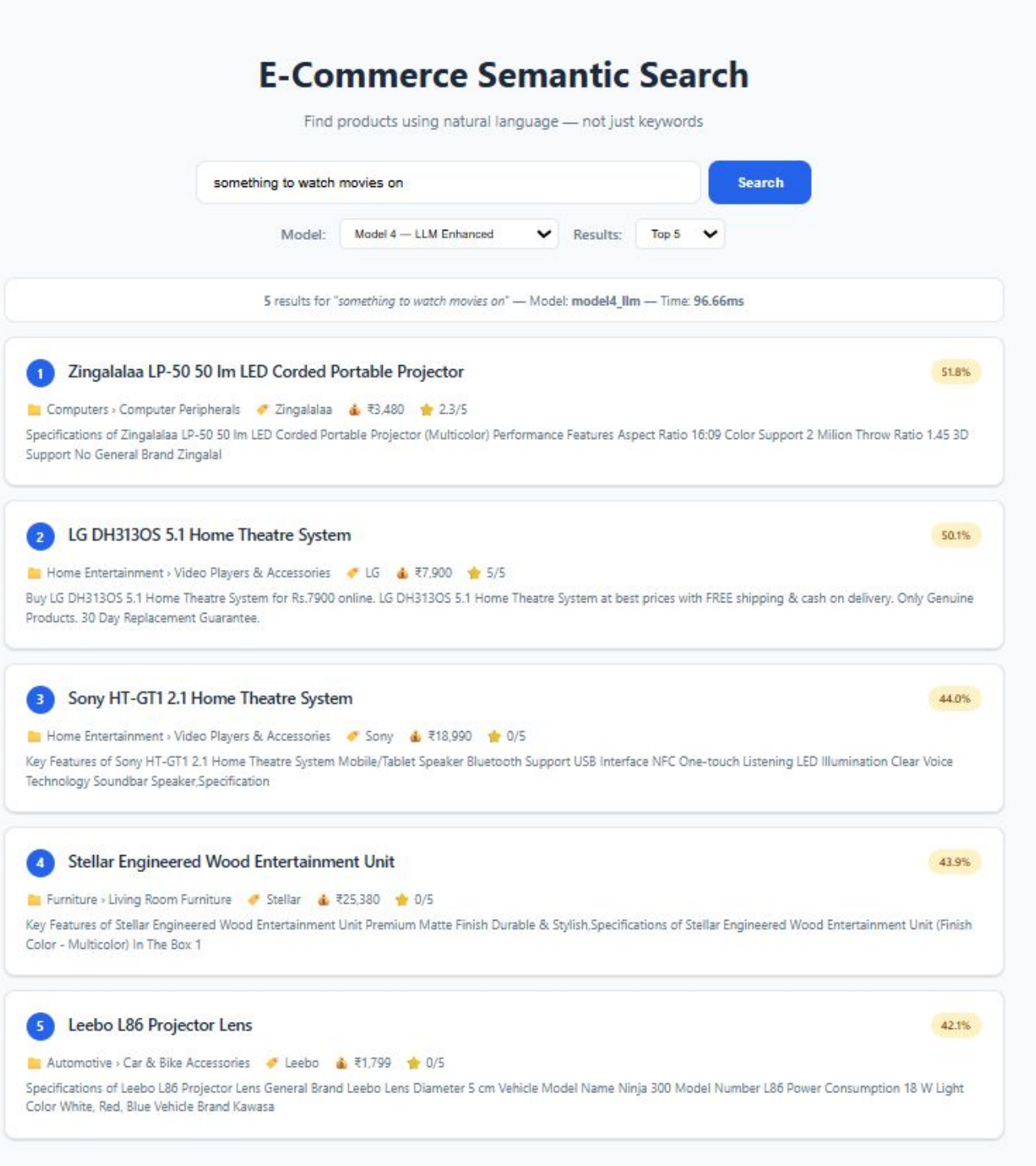
- Top results are still mostly home theatre systems, with better ranking (LG comes first due to high rating), showing metadata re-ranking is influencing results.
- One highly irrelevant item (stick-on bra) appears, indicating that richer embeddings can sometimes introduce noise from loosely related text.
- Some partially relevant items like entertainment unit and mobile stand are retrieved, showing broader understanding but weaker filtering.
- Overall, improves ranking quality compared to Model 2, but still struggles with vague intent and category precision.



QUERY RESULTS MODEL 4

Query: "something to watch movies on"

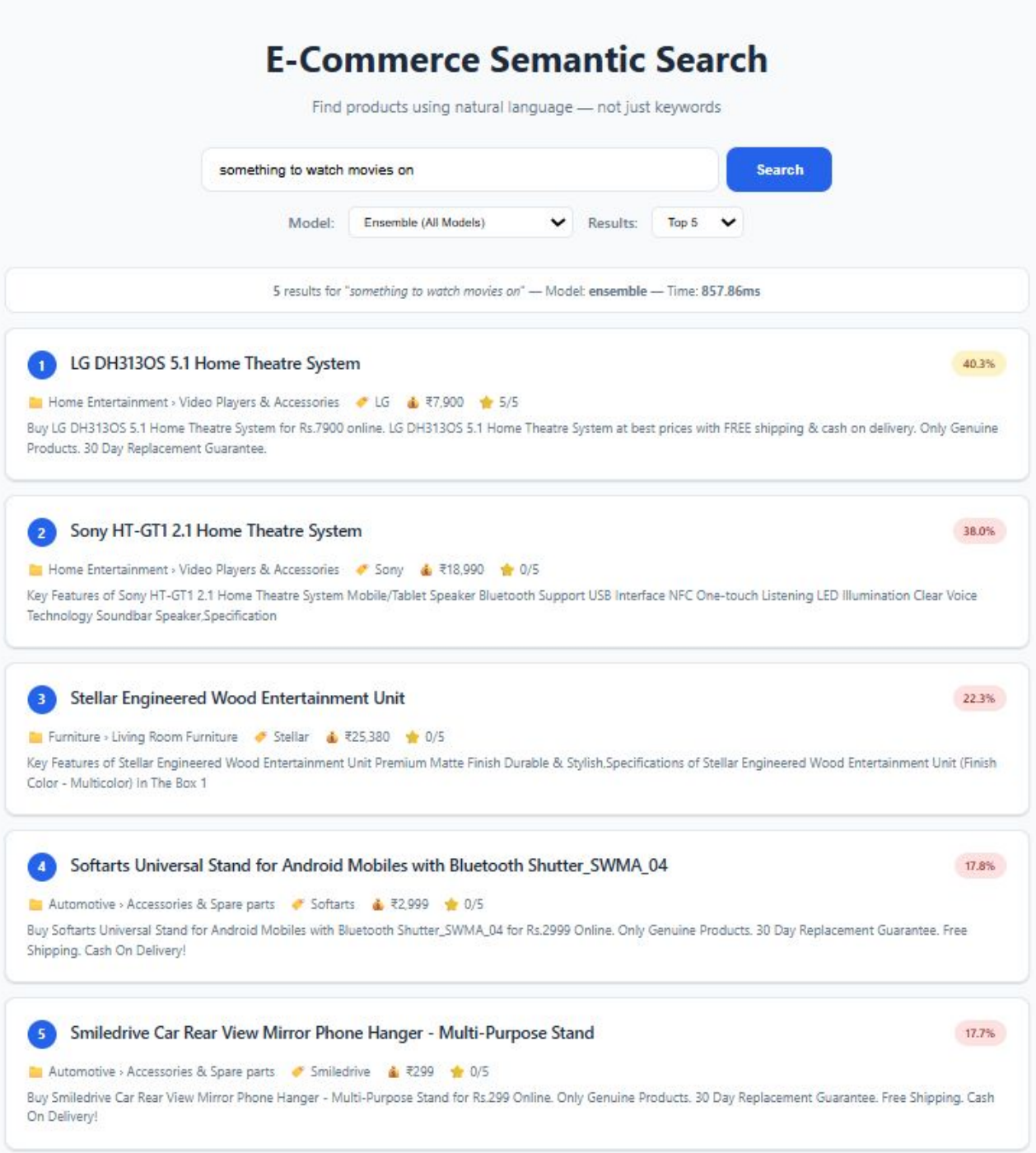
- Retrieves a projector as the top result, which directly matches the intent, showing successful query expansion from vague language to concrete products.
- Also returns home theatre systems, aligning well with ground truth categories.
- Still includes some loosely related items (entertainment unit, projector lens), but overall relevance is much better than previous models.
- Demonstrates strong understanding of user intent, making it most effective for vague or conversational queries.

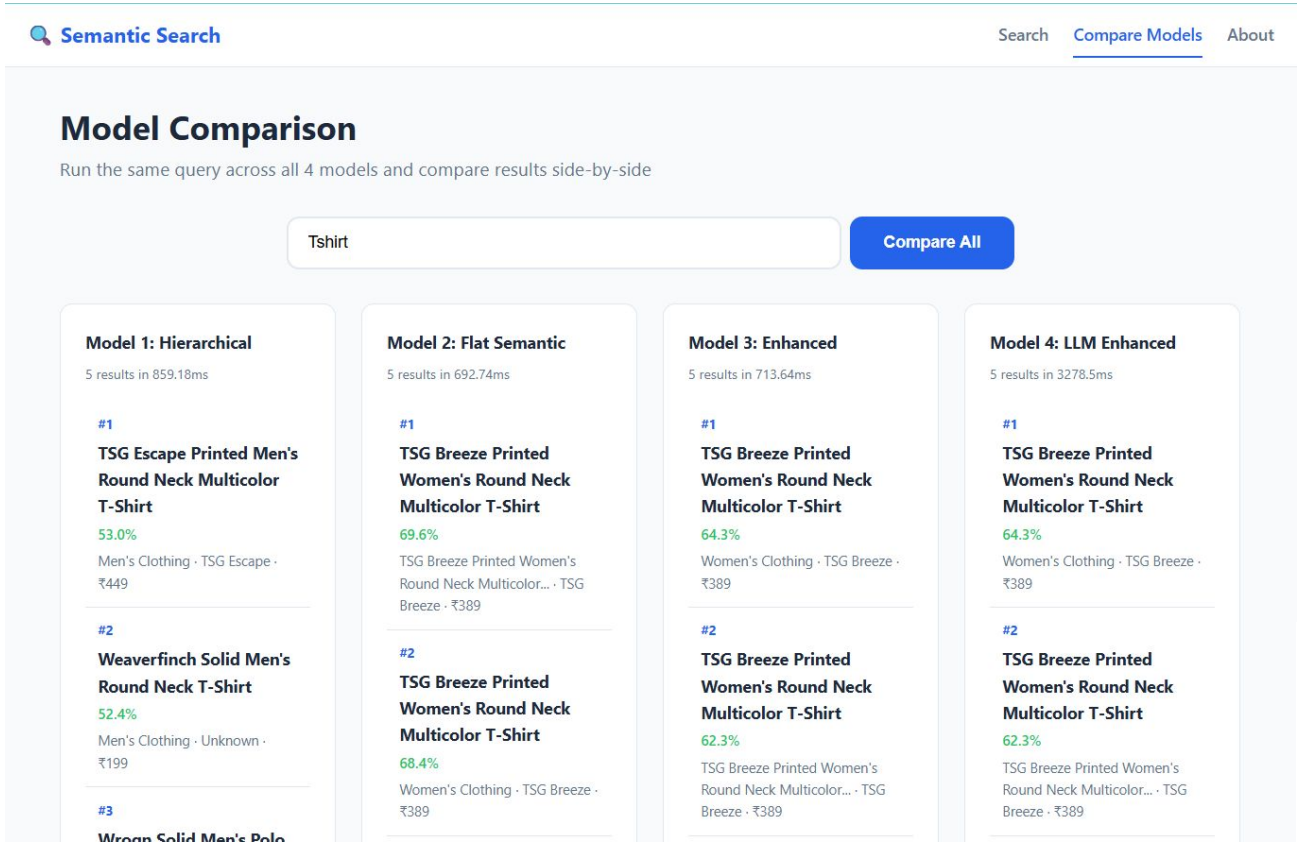


QUERY RESULTS ENSEMBLE

Query: "something to watch movies on"

- Top results are stable and relevant (home theatre systems), showing that combining models improves consistency and ranking quality.
- However, it misses strong hits like projectors (seen in Model 4), meaning ensemble can dilute some good signals from individual models.
- Lower-ranked results still include irrelevant items, so noise from weaker models is not fully removed.
- Overall, gives balanced and reliable results, but not always the most precise for vague queries compared to the LLM-enhanced model.





#3
Wrogn Solid Men's Polo Neck T-Shirt
52.1%
Men's Clothing · Unknown · ₹979

#4
Wrogn Solid Men's Round Neck T-Shirt
51.7%
Men's Clothing · Unknown · ₹989

#5
Wrogn Solid Men's V-neck T-Shirt
50.8%
Men's Clothing · Unknown · ₹449

Women's Clothing · TSG Breeze · ₹389

#3
LAST BENCH Printed Boy's Round Neck Blue T-Shirt
63.4%
Kids' Clothing · LAST BENCH · ₹249

#4
TSG Escape Printed Men's Round Neck Multicolor T-Shirt
61.2%
Men's Clothing · TSG Escape · ₹449

#5
TSG Breeze Casual Short Sleeve Embroidered Women's Top
58.5%
Women's Clothing · Unknown · ₹549

Round Neck Multicolor... · TSG Breeze · ₹389

#3
TSG Escape Printed Men's Round Neck Multicolor T-Shirt
57.8%
Men's Clothing · TSG Escape · ₹449

#4
TSG Breeze Casual Sleeveless, Short Sleeve Solid, Embellished Women's Top
57.6%
Women's Clothing · Unknown · ₹599

#5
TSG Breeze Casual Short Sleeve Embroidered Women's Top
57.4%
Women's Clothing · Unknown · ₹549

Round Neck Multicolor... · TSG Breeze · ₹389

#3
TSG Escape Printed Men's Round Neck Multicolor T-Shirt
57.8%
Men's Clothing · TSG Escape · ₹449

#4
TSG Breeze Casual Sleeveless, Short Sleeve Solid, Embellished Women's Top
57.6%
Women's Clothing · Unknown · ₹599

#5
TSG Breeze Casual Short Sleeve Embroidered Women's Top
57.4%
Women's Clothing · Unknown · ₹549

EVALUATION STRATEGY: SETUP

- The system was evaluated using a fixed benchmarking script that runs all models on the same set of test queries for fair comparison.

- We used **three types of queries**:
  - Exact : direct keyword match intent
  - Semantic : meaning-based queries
  - Vague : conversational or unclear user intent

- Each query is run against all models (Model 1–4) and their ensemble, and results are retrieved using **Top-K = 5 products per query**.

- We also included an **ensemble model**, which combines outputs from all individual models to compare whether merging improves ranking quality.

- For each query, the system checks whether retrieved products match expected **category-level ground truth terms** like product type, category, or subcategory.

- This setup ensures evaluation covers both structured search behavior (exact queries) and real-world user behavior (vague queries).

EVALUATION STRATEGY: METRICS

- The following metrics were used to measure the performance of the search engine:
  a. **Precision@K** measures how many of the top-K retrieved results are actually relevant, showing result accuracy.
  b. **Recall@K** measures how many relevant items were successfully retrieved within the top-K results, showing coverage.
  c. **MRR (Mean Reciprocal Rank)** checks how early the first correct result appears in the ranked list; higher means better ranking.
  d. **NDCG (Normalized Discounted Cumulative Gain)** evaluates ranking quality by giving higher weight to relevant results appearing at top positions.
  e. **F1 Score** combines precision and recall into a single balanced score for overall retrieval quality.

- Relevance is determined using **category-based matching**, where a result is marked correct if it matches expected (ground truth) product categories for each query as specified in the benchmarking script.
- Along with accuracy metrics, we also measure **average query latency (ms)** to evaluate speed vs quality trade-offs across models.

EVALUATION BENCHMARKS

Model	Type	P@K	R@K	MRR	NDCG	F1	Time
model1_hierarchical	exact	0.453	0.600	0.556	0.571	0.502	294.0ms
model1_hierarchical	semantic	0.723	0.846	0.795	0.813	0.769	80.5ms
model1_hierarchical	vague	0.629	0.857	0.720	0.774	0.694	117.2ms
model2_flat_semantic	exact	0.800	0.867	0.867	0.857	0.826	10.0ms
model2_flat_semantic	semantic	0.800	0.923	0.846	0.883	0.845	11.4ms
model2_flat_semantic	vague	0.614	0.857	0.714	0.759	0.691	11.7ms
model3_enhanced	exact	0.680	0.800	0.767	0.778	0.717	12.9ms
model3_enhanced	semantic	0.815	1.000	0.859	0.897	0.871	12.2ms
model3_enhanced	vague	0.471	0.786	0.702	0.722	0.543	13.1ms
model4_llm	exact	0.680	0.800	0.767	0.778	0.717	29.7ms
model4_llm	semantic	0.815	1.000	0.859	0.897	0.871	11.9ms
model4_llm	vague	0.586	0.929	0.702	0.765	0.675	44.2ms
ensemble	exact	0.760	0.867	0.833	0.844	0.802	87.6ms
ensemble	semantic	0.785	1.000	0.877	0.904	0.842	103.5ms
ensemble	vague	0.571	0.857	0.764	0.769	0.654	114.1ms

EVALUATION BENCHMARKS

Model Name	Query Type	Precision@5	NDCG	Avg Time (ms)
Model 1 (Hierarchical)	Semantic	0.723	0.813	205.2
Model 2 (Flat)	Exact	0.787	0.853	17.1
Model 3 (Enhanced)	Semantic	**0.815**	**0.897**	**16.7**
Model 4 (LLM)	Vague	0.629	0.801	2354.9
Ensemble	Exact	0.747	0.816	2544.8

Conclusion: Model 3 (Enhanced) provides the highest NDCG and fastest execution. Model 4 suffers from 2300ms+ API latency. The Ensemble maintains a high recall floor across all query forms.

Evaluation Insights

Model 1 (Hierarchical):
Performs better on semantic and vague queries due to category structure, but is slower and less precise for exact matches.

Model 2 (Flat Semantic):
Fastest model with strong performance on exact queries, but struggles with vague queries due to lack of contextual enrichment.

Model 3 (Enhanced Semantic) :
Best overall performance, especially on semantic queries (highest Precision@K and NDCG), showing the impact of enriched descriptions.

Model 4 (LLM-based):
Good semantic understanding but high latency (~2300ms) makes it impractical for real-time use.

Ensemble Model :
Provides balanced and robust performance across all query types, but introduces additional latency.

CONCLUSION & FUTURE WORK

By transitioning from key word base matching to continuous vector representations, we have successfully developed a search architecture that understands human intent.

Next Steps: Transitioning FAISS to a distributed cloud instance, refining the LLM prompts to reduce latency, and implementing live user-feedback analytics to dynamically adjust Model 3's reranking weights.

IMAGE SOURCES



https://static.vecteezy.com/system/resources/thumbnails/069/150/723/small/minimal-one-line-drawing-of-magnifying-glass-with-question-mark-search-icon-concept-in-black-and-white-line-art-vector.jpg

Source: www.vecteezy.com



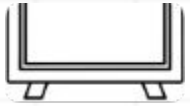
https://pnglove.com/data/img/1969_HEF1.jpg

Source: pnglove.com



https://static.vecteezy.com/system/resources/previews/070/097/524/non_2x/minimalist-black-and-white-brain-with-circuit-inspired-design-artificial-intelligence-and-neural-network-concept-illustration-vector.jpg

Source: www.vecteezy.com



https://static.vecteezy.com/system/resources/previews/065/683/303/non_2x/television-outline-design-modern-television-stand-minimalist-style-television-illustration-template-vector.jpg

Source: www.vecteezy.com